

Date : / / 20

SEAT NO.

201P III XX

(2829aa)

1 - 60

(E)

CHEMISTRY

Std. : 12th

Time : 3 Hours

(3 Pages)

Max. Marks : 70

General Instructions :

- 1) All the questions are compulsory.
- 2) Section 'A' contains Q. No. 1 to 4 of multiple choice type of questions carrying one mark each.
Q. No. 5 to 8 are very short answer type of questions carrying one mark each.
- 3) Section 'B' contains Q. No. 9 to 15 of short answer type of questions carrying two marks each.
Internal choice is provided to only one question.
- 4) Section 'C' contains Q. No. 16 to 26 of short answer type questions carrying three marks each.
Internal choice is provided to only one question.
- 5) Section 'D' contains Q. No. 27 to 29 of long answer type questions carrying five marks each.
Internal choice is provided to each question.
- 6) Use log table is allowed. Use of calculator is not allowed.
- 7) Given data - Gas constant $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$; Atomic Weight: $N = 14$; Avogadro's no. $(N_A) = 6.022 \times 10^{23}$

Section - A (1 Mark each)

(8)

Q.1 Which reagent cannot be used to prepare alkyl halide from an alcohol ?

- A) $\text{HCl} + \text{ZnCl}_2$ B) NaCl C) PCl_5 D) SOCl_2

Q.2 A matchbox exhibits

- A) Cubic geometry B) Monoclinic geometry
C) Orthorhombic geometry D) Tetragonal geometry

Q.3 A reaction showing slag formation is

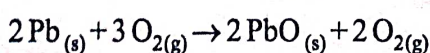
- A) $\text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow 6\text{Cu} + \text{SO}_2$ B) $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$
C) $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$ D) $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$

Q.4 Which is not an optically active amino acid ?

- A) Alanine B) Glycine C) Arginine D) Lysine

Q.5 At what temperature will the following reaction have $\Delta G = -777.8 \text{ kJ}$, if $\Delta H = -843.7 \text{ kJ}$ and

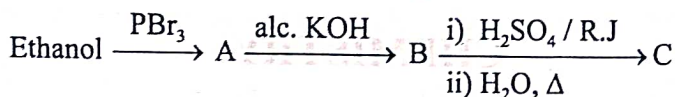
$\Delta S = -165 \text{ kJ}$



Q.6 For a general chemical reaction, $mA + nB \rightarrow pC + qD$. The rate law is $\frac{dx}{dt} = K[A]^x[B]^y$. 1

What is the molecularity and overall order of reaction?

Q.7 Identify 'C' in the following reaction. 1



Q.8 Arrange the following amines in increasing order of basicity in aqueous solution. 1

- i) CH_3NH_2 ii) $(\text{CH}_3)_2\text{NH}$ iii) $(\text{CH}_3)_3\text{N}$

Section - B (2 Mark each)

(14)

Q.9 Draw a neat and labelled diagram of Bessemer converter used in extraction of copper. 2

Q.10 Classify the following ligands into monodentate, polydentate and ambidentate. 2

- i) NH_3 ii) ethylene diamine iii) NO_2^- iv) oxalato

OR

Q.10 Draw the geometrical isomers of $[\text{CoCl}_2(\text{en})_2]^+$. 2

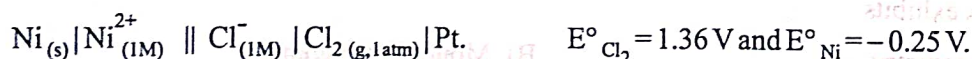
Q.11 The solubility of dissolved oxygen to 27°C is $2.6 \times 10^{-3} \text{ mol dm}^{-3}$ at 2 atm. Find its solubility at 8.4 atm. and 27°C . 2

Q.12 How is nylon - 66 prepared? 2

Q.13 Draw the structure of $\text{H}_4\text{P}_2\text{O}_6$ and $\text{H}_4\text{P}_2\text{O}_7$. Find out oxidation state of phosphorus in each oxyacid. 2

Q.14 Write ozonolysis reaction for propylene. 2

Q.15 Write the cell reaction and calculate the standard potential of the cell. 2



Section - C (3 Mark each)

(33)

Q.16 State Kohlrausch's law. Distinguish between electronic and electrolytic conductors. 3

Q.17 Define - Lipids. How are lipids classified? 3

Q.18 What is freezing point of a liquid? How is molar mass of a non-volatile solute related to the depression in freezing point? Derive an equation for it. 3

Q.19 State the hybridization of the nitrogen atom in amines. Write a short note on Mendius reduction. 3

Q.20 What is the action of acidified potassium dichromate on - 3

- a) SO_2 b) KI c) ethanol?

Q. 21 The rate constant of a first order reaction is $6.8 \times 10^{-4} \text{ s}^{-1}$. If the initial concentration of the reactant is 0.04 M, what is its molarity after 20 minutes? How long will it take for 25% of the reactant to react? 3

Q. 22 What are broad spectrum antibiotics? Explain any two chemical methods of food preservation. 3

Q. 23 Predict the expected product of substitution reactions. 3

a) Isobutyl chloride + Sodium ethoxide

b) n-butyl chloride + sodium

c) 1-chloropropane + aq. Potassium hydroxide

Q. 24 Explain how does nitrogen exhibit anomalous behaviour amongst gr. 15 elements. 3

Q. 25 How is diethyl ether prepared by continuous etherification process? Why is the process so called? 3

OR

Q. 25 How is carbolic acid prepared from cumene? What is the action of bromine water on carbolic acid? 3

Q. 26 2.8×10^{-2} kg of nitrogen is expanded isothermally and reversibly at 300 K from $15.15 \times 10^5 \text{ Nm}^{-2}$ when the work done is found to be -17.33 kJ . Find the final pressure. 3

Section - D (5 marks each)

(15)

Q. 27 Why does copper shows abnormal electronic configuration? 2

An element crystallises in fcc crystal lattice and has a density of 10 g cm^{-3} with unit cell edge length of 100 pm. Calculate number of atoms present in 1 g of crystal. 3

OR

Q. 27 What are chemical twins? Give examples. 2

A metal crystallises into two cubic faces namely face centered (fcc) and body centered (bcc) whose unit cell edge lengths are $3.5\text{\AA}$ and $3\text{\AA}$ respectively. Find the ratio of the densities of fcc and bcc. 3

Q. 28 How is orlon prepared? 1

Write the formula of the complex Diamminedichloridoplatinum (II). 1

Explain the structure of SO_2 molecule. Write uses of SO_2 . 3

OR

Q. 28 How is teflon prepared? 1

Write IUPAC name for the complex : $\text{Na}_2[\text{Zn}(\text{CN})_4]$ 1

What is the action of chlorine on - a) hydrogen b) turpentine oil c) CS_2 ? 3

Q. 29 State and explain Hess's law of constant heat summation.

Write a note on : Aldol condensation.

OR

Q. 29 Which of the following pairs have larger entropy ? Why ?

- a) $\text{CO}_{2(s)}$ or $\text{CO}_{2(g)}$ b) $\text{CH}_3\text{OH}_{(l)}$ or $\text{C}_2\text{H}_5\text{-O-C}_2\text{H}_5_{(l)}$

How will you bring about the following conversions ?

- a) Ethylidene dichloride to acetaldehyde.
b) Calcium acetate to acetone.
c) Cyclopropane carboxylic acid to cyclopropyl methanol.